Power Budget Example

Team Number:	**Team 307**
Project Name:	**Submersible Exploration Device**
Team Member Names:	**Abriana Poola**

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators,

All Major Components	**Component Name**	**Part Number**	**yVoltageRar**	**#**	**aximumCurre**	**alCurrent(mA)**	**Unit**
	ESP32-S3-WROOM-1	ESP32-S3-WR	3.0V - 3.6V	1	500	500	mA
	OLED display module	B085WCRS7C	3.3V	1	25	25	mA
	LED 1	Generic LED	3.3V	1	6	6	mA
	LED 2	Generic LED	3.3V	1	6	6	mA
							mA
							mA

B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in

+12V Power Rail	**Component Name**	**Part Number**	**yVoltageRar**	**#**	**aximumCurre**	**alCurrent(mA)**	**Unit**
						0	mA
						0	mA
						0	mA
						0	mA
						0	mA
					Subtotal	0	mA
					Safety Margin	25%	
					Total Current Required on +12V Rail	0	mA
c1. Regulator or Source Ch	Direct +12V input from wa	Barrel jack J3	12V nominal	1	1000	1000	mA
					Total Remaining Current Available on +12V Rail	1000	mA

+3.3V Power Rail	**Component Name**	**Part Number**	**yVoltageRar**	**#**	**aximumCurre**	**alCurrent(mA)**	**Unit**
	ESP32-S3-WROOM-1	ESP32-S3-WR	3.0V - 3.6V	1	500	500	mA
	OLED display module	B085WCRS7C	3.3V	1	25	25	mA
	LED 1	Generic LED	3.3V	1	6	6	mA
	LED 2	Generic LED	3.3V	1	6	6	mA
					Subtotal	537	mA
					Safety Margin	25%	
					Total Current Required on +3.3V Rail	671.25	mA
c2. Regulator or Source Ch	+3.3V buck regulator	LM2576HVS-3	4V - 60V inpu	1	3000	3000	mA
					Total Remaining Current Available on 3.3V Rail	2328.75	mA

D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of

External Power Source 1	**Component Name**	**Part Number**	**yVoltageRar**	**Outpu**	**aximumCurre**	**alCurrent(mA)**	**Unit**
Power Source 1 Selection	12V wall adapter	B0D269PYK9	AC input	12V	1000	1000	mA
Power Rails Connected to External Power Source 1	+3.3V buck regulator inpu	LM2576HVS-3	5m 12V sour	1	231	231	mA
						0	mA
						0	mA
					Total Remaining Current Available on External Power Source 1	769	mA

External Power Source 2	**Component Name**	**Part Number**	**yVoltageRar**	**Outpu**	**aximumCurre**	**alCurrent(mA)**	**Unit**
Power Source 2 Selection	Not used	N/A	N/A	N/A	0	0	mA
Power Rails Connected to External Power Source 2						0	mA
						0	
						0	
					Total Remaining Current Available on External Power Source 2	0	mA
					Battery Life		hours

Notes / Assumptions

ESP32 budget uses 500 mA peak design current so the 3.3 V rail can handle Wi-Fi startup/current spikes.
OLED current estimated at 25 mA because the schematic does not include the OLED module part number.
LED current assumes 220Ω series resistors and ~2.0V LED forward drop: (3.3V - 2.0V)/220Ω ≈ 5.9mA each.
External 12V source current is estimated from 3.3V rail load using 80% buck efficiency.